

Cointegration

Uncovering long-run relationships in time series

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November 7, 2025



① Pre-requisites

② Why?

③ What?

④ How?

⑤ References

① Pre-requisites

② Why?

③ What?

④ How?

⑤ References

Prerequisites

Time Series

Examples I

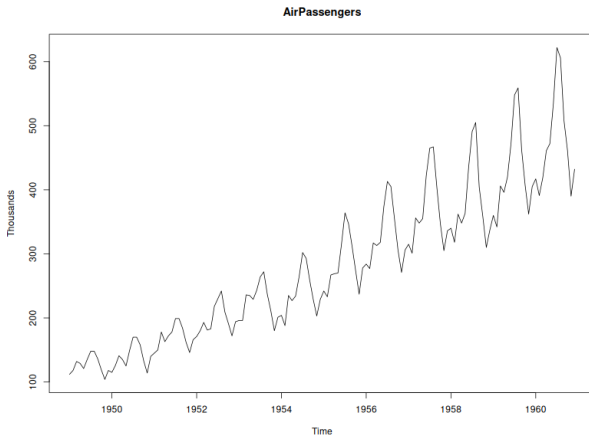


Figure 1: Example of a **non-stationary** time series

Examples II

Refer: Monthly airline passengers (1949-1960) [R-datasets]

Examples III

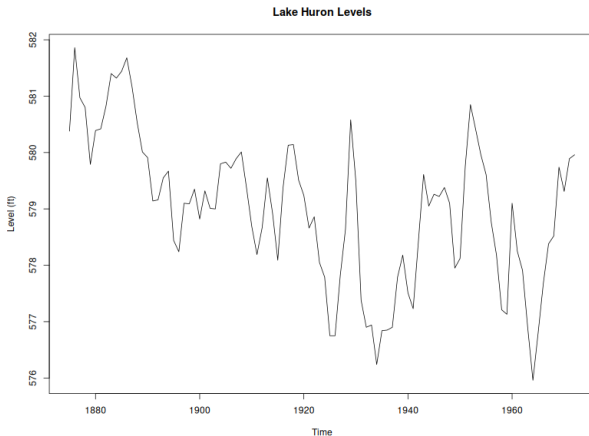


Figure 2: Example of a **nearly stationary** time series

Examples IV

Refer: Annual level of Lake Huron (1875-1972) [R-datasets]

Examples V

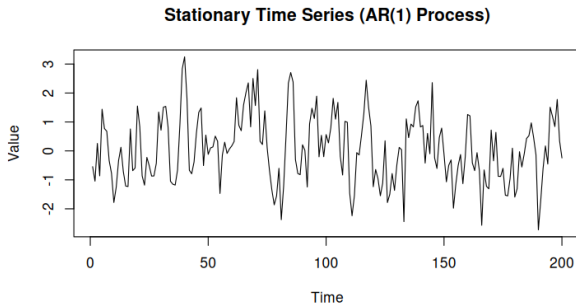


Figure 3: Example of a **stationary** time series

It is AR(1) process.

Pre-requisites I

What is a stationary time series?

A stationary time series is one whose statistical properties do not change over time. Formally, a time series y_t is weakly stationary if:

- Constant mean: $E[y_t] = \mu \forall t$
- $E[y_t^2] < \infty$
- Covariance depends only on lag: $\text{Cov}(y_t, y_{t-k}) = \gamma_k$, which only depends on the lag k and not on time t .

Pre-requisites II

Unit root geometrically

- A unit root means one of these roots lie on the unit circle, i.e., $|z| = 1$. Eg: $y_t = \phi y_{t-1} + a_t$.
- Characteristic equation: $1 - \phi B = 0 \implies B = \frac{1}{\phi}$.
 - If $|\phi| < 1$, then the root lies outside the unit circle $\implies$ stationary.
 - If $|\phi| = 1$, then the root lies exactly on the unit circle $\implies$ unit root non-stationary.
 - If $|\phi| > 1$, then the root lies outside the unit circle $\implies$ explosive series.

Pre-requisites III

Property	Stationary	Unit Root (Nonstationary)	Differenced (Stationary)
Equation	$y_t = 0.5y_{t-1} + a_t$	$y_t = y_{t-1} + a_t$	$\Delta y_t = a_t$
Process Type	AR(1), $I(0)$	Random Walk, $I(1)$	White Noise, $I(0)$
Stationarity	Yes	No	Yes

Table 1: Comparison of stationary, unit root, and differenced time series.

① Pre-requisites

② Why?

Problem

Trap

③ What?

④ How?

⑤ References

Problem

How do we test for unit roots? Goal?

Unit roots testing I

Aim: Check if the data is $I(0)$ or $I(1)$.

Test: Augmented Dickey-Fuller (ADF) test

Method: Run the following regression:

$$\Delta z_t = \beta z_{t-1} + \sum \phi_i^* \Delta z_{t-i} + a_t$$

Unit roots testing II

Hypothesis testing

- $H_0 : \beta = 0 \rightarrow \exists$ a unit root ($I(1)$)
- $H_1 : \beta < 0 \rightarrow$ stationary ($I(0)$)

① Pre-requisites

② Why?

Problem

Trap

③ What?

④ How?

⑤ References

Problem

Most real-world data are non-stationary.

① Pre-requisites

② Why?

Problem

Trap

③ What?

④ How?

⑤ References

The trap I

Two Independent Random Walks (I(1))

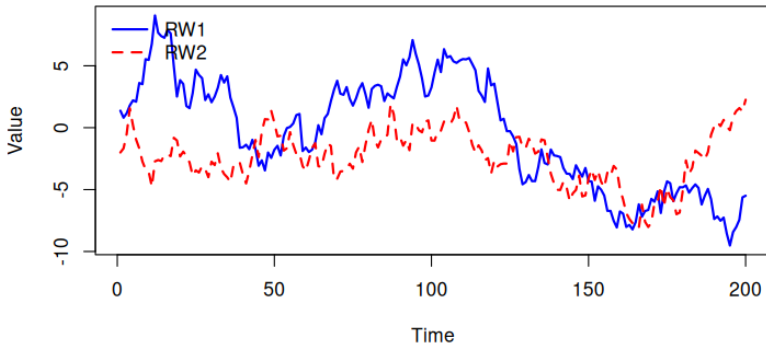


Figure 4: Two random walks

The trap II

- A **regression** may show a high R^2 and significant t-statistics, even though there is no real relationship between the variables.
- The regression is ‘**spurious**’ or ‘**fake**’. It captures the common trends, not a true economic relationship. This is a major pitfall.

The trap III

Example:

Suppose

- X_t = global temperature over time
- Y_t = stock prices over time

Both may increase over decades, so a regression explaining the phenomena would be:

$$Y_t = \alpha + \beta X_t + \epsilon_t$$

might show a very high R^2 and a significant β due to its shared trend, not because of the variables affecting each other. But **temperatures** & **stock prices** do not affect each other in real-world scenario. Hence this type of regression is called **spurious regression**.

Correlation vs Cointegration I

Feature	Correlation	Cointegration
Time frame	Short-term link	Long-term link
Type of data	Works with any data	Needs non-stationary series
Focus	Linear co-movement	Long-run equilibrium
Nature	Simple dependence	Complex, involves stationarity
Example	Two stocks move together	Oil and gas prices linked

Table 2: Correlation vs. Cointegration comparison

① Pre-requisites

② Why?

③ What?

Definition

Example

Method

④ How?

⑤ References

1 Pre-requisites

2 Why?

3 What?

Definition

Example

Method

4 How?

5 References

Co-integration I

What is **Co-integration**?

Co-integration II

Definition: Two time series X_t, Y_t are cointegrated if:

- Each series is **integrated of order 1** $[I(1)] \rightarrow$ they become stationary after first differencing.
- $\exists$ a linear combination, say $u_t = Y_t - \beta X_t$, which becomes **stationary** $[I(0)]$.

Co-integration III

- A vector of k time series is said to be **cointegrated** if *individual components are non-stationary* but $\exists$ a linear combination β such that $w_t = \beta' z_t$ is *stationary*.
- If there are m such vectors linearly independent, it means that $\exists m$ cointegrating relationships and $k-m$ sources of non-stationarity.

Co-integration IV

Examples

- Purchasing power parity.
- Pair trading in stock market.
- Income and consumption.

Role of cointegration

Cointegration is about long-run equilibrium relationships between non-stationary $I(1)$ variables.

- If two or more series share a *common stochastic trend*, then some **linear combination** of them will be **stationary $I(0)$** .

$$y_t \sim I(1), x_t \sim I(1) \text{ but } (y_t - \beta x_t) \sim I(0)$$

- Cointegration does not depend on whether each individual time series is AR, MA or ARMA. Their order of integration matters.

1 Pre-requisites

2 Why?

3 What?

Definition

Example

Method

4 How?

5 References

Example I

Consider a **bivariate (k=2) VARMA(1,1)**

$$\begin{bmatrix} z_{1t} \\ z_{2t} \end{bmatrix} - \begin{bmatrix} 0.5 & -1.0 \\ -0.25 & 0.5 \end{bmatrix} \begin{bmatrix} z_{1,t-1} \\ z_{2,t-1} \end{bmatrix} = \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix} - \begin{bmatrix} 0.2 & -0.4 \\ -0.1 & 0.2 \end{bmatrix} \begin{bmatrix} a_{1,t-1} \\ a_{2,t-1} \end{bmatrix}$$

This is in the form $\Phi(B)z_t = \Theta(B)a_t$, where $\Phi(B)$ is the AR coefficient matrix & $\Theta(B)$ is the white noise coefficient matrix.

Example II

The system is nonstationary

Observations

- The AR coefficient matrix Φ has eigenvalues of 0 and 1.
- The eigenvalue of 1 causes non-stationarity.
- The eigenvectors are $(1.0, -2.0)^t$ & $(0.5, 1.0)^t$ corresponding to the eigenvalues 0 & 1 respectively.

Example III

We apply a special transformation matrix L to create a new system

y_t :

$$\begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} 1.0 & -2.0 \\ 0.5 & 1.0 \end{bmatrix} \begin{bmatrix} z_{1t} \\ z_{2t} \end{bmatrix} \quad \text{or} \quad y_t = Lz_t$$

Example IV

So, $L =$

$$\begin{bmatrix} 1.0 & -2.0 \\ 0.5 & 1.0 \end{bmatrix}$$

and let $y_t = Lz_t$.

So,

$$y_{1t} = 1.0z_{1t} - 2.0z_{2t}$$

$$y_{2t} = 0.5z_{1t} + 1.0z_{2t}$$

- $y_{1t} \implies$ common trend due to $I(1)$ non-stationarity.
- $y_{2t} \implies$ white noise due to $I(0)$ stationarity.

① Pre-requisites

② Why?

③ What?

Definition

Example

Method

④ How?

⑤ References

Error Correction Model (ECM) I

- *Engle-Granger* suggested the **ECM form**. This is a different algebraic representation for the same VARMA model that is **stationary** and **invertible**.
- ECM form of a VARMA(p,q) model (VECM itself):

$$\Delta z_t = \Pi z_{t-1} + \sum_{i=1}^{p-1} \Phi_i^* \Delta z_{t-i} + c(t) + \Theta(B)a_t$$

- For univariate case, it is simply ECM; for *multivariate* cases, it is **VECM (Vector Error Correction Model)**

Error Correction Model (ECM) II

VECM form of VARMA(p,q):

$$\Delta z_t = \Pi z_{t-1} + \sum_{i=1}^{p-1} \Phi_i^* \Delta z_{t-i} + c(t) + \Theta(B)a_t$$

where

- 1 $\sum_{i=1}^{p-1} \Phi_i^* \Delta z_{t-i}$ captures the short-run dynamics.
- 2 $c(t) + \Theta(B)a_t$ is the stationary part.
- 3 Πz_{t-1} is the error correction term.

VECM I

For **VAR(2)** model,

$$z_t = c_0 + \phi_1 z_{t-1} + \phi_2 z_{t-2} + a_t$$

To make it stationary, we difference it once,

$$z_t - z_{t-1} = c_0 + \phi_1 z_{t-1} - z_{t-1} + \phi_2 z_{t-2} + a_t$$

$$\implies \Delta z_t = c_0 + (\phi_1 - I)z_{t-1} + \phi_2 z_{t-2} + a_t$$

$$\implies \Delta z_t = c_0 + \Pi z_{t-1} + \Phi_1^* \Delta z_{t-1} + a_t$$

where

$$\Pi = (\phi_1 + \phi_2 - I), \quad \Phi_1^* = -\phi_2$$

VECM II

In general, **VECM** model looks like

$$\Delta z_t = c(t) + \Pi z_{t-1} + \sum_{i=1}^{p-1} \Phi_i^* \Delta z_{t-i} + a_t$$

where

$$\Pi = -(I - \sum_{i=1}^p \phi_i) \text{ \& } \Phi_i^* = -(\sum_{j=i+1}^p \phi_j)$$

① Pre-requisites

② Why?

③ What?

④ How?

⑤ References

Johansen test I

- **Goal:** We need a formal test to find out if cointegration exists and, if so, how many (i.e., how many “leashes”).
- **Test:** The Johansen test (Section 5.9) is the standard tool.
- **The model:** Vector Error-Correction Model (VECM) form (from Section 5.6):

$$\Delta z_t = \Pi z_{t-1} + \sum_{i=1}^{p-1} \Phi_i^* \Delta z_{t-i} + c(t) + \Theta(B)a_t$$

- **Objective:** To check for the rank (m) of the Π matrix which is the number of cointegrating vectors.
 - If $\text{Rank}(\Pi) = 0$, the series are not cointegrated.
 - If $\text{Rank}(\Pi) = m > 0$, there are m cointegrating vectors.

Johansen test II

- **Test statistics:** The test produces two statistics (Section 5.9.4):

- ① Trace statistic (L_{tr}): Tests

$$H_0 : \text{Rank} = m \text{ vs. } H_a : \text{Rank} > m$$

- ② Max-Eigenvalue statistic (L_{max}): Tests

$$H_0 : \text{Rank} = m \text{ vs. } H_a : \text{Rank} = m + 1$$

- $L_{tr}(m) = -(T - kp) \sum_{i=m+1}^k \ln(1 - \lambda_i)$
- $L_{max}(m) = -(T - kp) \ln(1 - \lambda_{m+1})$
- We keep testing sequentially for all values of m until it reaches $\text{Rank}(\Pi)$.

Example I

Setup: We have two nonstationary ($I(1)$) series $z_t = (z_{1t}, z_{2t})'$, and we wish to test for the number of cointegrating relationships.

Step 1: Set up the ECM Model Choose a VAR order $p = 2$. Based on the text, the Error-Correction Model (ECM) is:

$$\Delta z_t = c_0 + \Pi z_{t-1} + \Phi_1^* \Delta z_{t-1} + a_t$$

Our goal is to determine the *rank* (m) of the 2×2 matrix Π :

- $m = 0$: no cointegration,
- $m = 1$: one cointegrating vector,
- $m = 2$: both series are stationary.

Example II

Step 2: Run “Helper” Regressions We cannot test Π directly. We first concentrate out short-run dynamics:

- 1 Regress Δz_t on a constant and Δz_{t-1} ; save residuals $\hat{u}_t$.
- 2 Regress z_{t-1} on a constant and Δz_{t-1} ; save residuals $\hat{v}_t$.

Example III

Step 3: Compute Covariance Matrices From the residuals, we compute sample covariance matrices (hypothetical values):

$$\hat{\Sigma}_{00} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}, \quad \hat{\Sigma}_{11} = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}, \quad \hat{\Sigma}_{01} = \begin{bmatrix} 3 & 1.5 \\ 1 & 1 \end{bmatrix}, \quad \hat{\Sigma}_{10} = \hat{\Sigma}'_{01} = \begin{bmatrix} 3 & 1 \\ 1.5 & 1 \end{bmatrix}$$

Example IV

Step 4: Solve the Eigenvalue Problem The Johansen test uses the eigenvalues of

$$S = \hat{\Sigma}_{11}^{-1} \hat{\Sigma}_{10} \hat{\Sigma}_{00}^{-1} \hat{\Sigma}_{01}.$$

Example V

Calculations:

- ① Find inverses:

$$|\hat{\Sigma}_{00}| = 11, \quad \hat{\Sigma}_{00}^{-1} = \frac{1}{11} \begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix}, \quad |\hat{\Sigma}_{11}| = 6, \quad \hat{\Sigma}_{11}^{-1} = \frac{1}{6} \begin{bmatrix} 2 & -2 \\ -2 & 5 \end{bmatrix}$$

- ② Multiply to obtain:

$$S = \begin{bmatrix} 0.3636 & 0.1591 \\ 0.2273 & 0.1932 \end{bmatrix}$$

- ③ Eigenvalues of S :

$$\lambda_1 \approx 0.487, \quad \lambda_2 \approx 0.070$$

Example VI

Step 5: Compute Test Statistics

Effective sample size: $T - kp = 196$.

Trace Statistic:

$$\begin{aligned}L_{\text{tr}}(0) &= -196 [\ln(1 - 0.487) + \ln(1 - 0.070)] \\ &= -196[-0.667 - 0.072] \approx \mathbf{144.8}\end{aligned}$$

$$L_{\text{tr}}(1) = -196 \ln(1 - 0.070) = -196[-0.072] \approx \mathbf{14.1}$$

Max-Eigenvalue Statistic:

$$L_{\text{max}}(0) = -196 \ln(1 - 0.487) = -196[-0.667] \approx \mathbf{130.7}$$

$$L_{\text{max}}(1) = -196 \ln(1 - 0.070) = -196[-0.072] \approx \mathbf{14.1}$$

Example VII

Step 6: Compare with Critical Values

Hypothesis	Statistic	Value	5% CV	Decision
<i>Trace Test</i>				
$H_0 : m = 0$	$L_{\text{tr}}(0)$	144.8	17.95	Reject H_0
$H_0 : m \leq 1$	$L_{\text{tr}}(1)$	14.1	8.18	Reject H_0
<i>Max-Eigen Test</i>				
$H_0 : m = 0$	$L_{\text{max}}(0)$	130.7	14.90	Reject H_0
$H_0 : m = 1$	$L_{\text{max}}(1)$	14.1	8.18	Reject H_0

Example VIII

Conclusion Since both tests reject H_0 for $m = 0$ and $m = 1$, we conclude that the rank is $m = 2$. Therefore, the original bivariate system z_t is stationary ($I(0)$) and has full rank.

① Pre-requisites

② Why?

③ What?

④ How?

⑤ References

References

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Thank You